SURYA NARAYANAN
Python Developer

SUMMARY:

- Over **4+ years of hands-on experience in Python backend development**, enterprise automation, and ERP system customization. Specialized in building scalable backend components, integrating complex workflows, and developing custom applications using **Python, Frappe, and ERPNext** to support domains such as Procurement, HR, Inventory, and Geospatial Data Management.

- Designed AI powered applications integrating **LLMs**, and **RAG pipelines**. Utilized embeddings, vector DBs and OCR pipelines for intelligent document processings.

- Experience in **FastAPI and Frappe** into production and scaling applications beyond single instances and understanding of cloud deployments to build applications.

- Leveraged **AWS services** for cloud-native architecture, ensuring scalability, high availability, and efficient resource management across deployed applications.

- Skilled in **Pandas, NumPy**, and Python-based data processing for ETL workflows, data transformation, report generation, and business analytics across ERP modules. Developed automated reports, dashboards, and analytical views to support data-driven decision-making.

- Detail-oriented and efficient, with a strong aptitude for project and time management, ensuring high-quality deliverables within specified timelines.

- Strong analytical, problem-solving, and communication skills with a proven ability to collaborate with clients, cross-functional teams, and end-users. Experienced working in **Agile environments**, following best practices in coding, debugging, testing, and iterative development. Comfortable working independently or as part of a collaborative engineering team.

- Designed and implemented automated workflows to integrate multiple APIs, services, and databases, reducing manual tasks and increasing operational efficiency.

TECHNICAL SKILLS:

LANGUAGES: Python, HTML, CSS, Javascript

WEB DEVELOPMENT:

- **Frameworks & Libraries:** FastAPI, Frappe - ERPNext, Pandas, Numpy
- **API & Protocols:** REST, OAuth2

DATABASES: PostgreSQL, MySQL, MariaDB, SQL

STORAGE: S3

VECTOR STORES: ChromaDB, Pinecone

CROSS-PLATFORM & ORCHESTRATION: Docker, Celery, Redis

DEVOPS & DEPLOYMENT: Docker, GitHub Actions

CLOUD PLATFORMS:

- **AWS:** S3

PROJECT MANAGEMENT & VERSION CONTROL: Git, GitHub, Jira, Trello, Linear

PROFESSIONAL EXPERIENCE:

Python Developer

Project Name: ERPNext

Technologies: Frappe, Python, HTML, CSS, JavaScript, MariaDB, Docker, Pandas, NumPy, AWS S3

Responsibilities:

1. **ERP Custom App Development & Workflow Automation**:
    - Designed and developed a custom Frappe-based ERP application on top of ERPNext to automate and streamline core business workflows across Procurement, Stock, and HR modules.
    - Implemented domain-specific logic, server scripts, custom DocTypes, API integrations, and workflow rules.
2. **Module & Doctype Enhancements**:
    - Maintained, customized, and extended existing ERPNext Doctypes, workflows, and permission structures to support evolving business requirements.
    - Ensured data integrity, compliance, and secure multi-level access control across modules.
3. **Advanced Reporting & Data Processing**:
    - Built complex analytical reports using **Pandas** and **NumPy** for data cleaning, transformation, aggregation, and KPI computation.
    - Enabled stakeholders to make data-driven decisions through accurate and automated reporting pipelines.
4. **Business Process Automation**:
    - Automated repetitive operations such as employee onboarding flows, scheduled background jobs, follow-up reminders, and email/SMS notifications, significantly reducing manual workload and improving process efficiency.

Description: ERPNext is an enterprise-grade open-source ERP platform designed to manage end-to-end business operations. This project involved developing a customized ERP system tailored to organizational workflows across modules like Procurement, Inventory, HR, and Administration. Through automation, data analytics, and seamless process integration, the solution improved operational efficiency, enhanced cross-department collaboration, and enabled real-time business insights.

Project Name: Strategy7

Technologies: Frappe, Python, HTML, CSS, JavaScript, MariaDB, Docker, Pandas, NumPy, AWS S3

Responsibilities:

1. **Geospatial API Integration & Data Sync**:
    - Developed secure API integrations to fetch and manage cadastral parcel and geospatial data from external Geoservers using authenticated HTTP requests.
    - Enabled seamless synchronization of GIS layers directly within the ERP system.
2. **Bulk Processing & Background Jobs**:
    - Implemented asynchronous bulk operations using Frappe background jobs for mass geospatial updates, layer generation, and data imports ensuring high performance and scalability when handling large datasets.
3. **Dynamic UI & Data Management Interfaces**:
    - Designed and developed user-friendly Frappe interfaces for managing cadastral parcels and enterprise records.
    - Implemented advanced filtering, sorting, search, and real-time data updates to improve usability and data accessibility.
4. **Role-Based Access & Security**:
    - Customized ERPNext's role and permission framework to enforce secure, hierarchical access to sensitive geospatial and enterprise data.
    - Ensured compliance, data protection, and controlled visibility across departments.
5. **Serverless & Event-Driven Features:**
    - AWS Lambda jobs scheduled to refresh embeddings from updated documents.
    - EventBridge triggers Slack/Email notifications for new knowledge updates.
6. **Data Analytics & Reporting**:
    - Utilized **Pandas** and **NumPy** to perform complex data transformations, aggregations, and report generation. Delivered actionable analytics for geospatial insights, enterprise performance, and cadastral parcel mapping.

Description: Strategy ERPNext is a custom extension built on top of ERPNext to support **cadastral parcel management, geospatial data integration, and enterprise mapping workflows**. The system bridges ERPNext with external Geoservers, enabling organizations to visualize, manage, and maintain geospatial layers alongside operational data. By combining automation, GIS synchronization, and structured data workflows, the application enhances workflow efficiency, improves decision-making, and supports seamless collaboration across teams handling geospatial and enterprise datasets.

Project Name: Teamicate

Technologies: Python, Django, Celery, Pandas, NumPy, AWS EC2, S3

Responsibilities:

1. **Intelligent Auto-Scheduling Engine**:
   - Developed a smart auto-pick scheduling algorithm using **Python** and **Celery** to automatically analyze participant availability and determine optimal meeting times, reducing manual coordination efforts.
2. **Calendar & Video Platform Integrations**:
   - Integrated Teamicate with third-party services including **Google Calendar, Apple Calendar, Zoom, and Google Meet**, enabling automated event creation, meeting confirmation, and seamless data synchronization across devices.
3. **Automation & Workflow Optimization**:
   - Implemented backend logic to automatically confirm, finalize, and notify participants about scheduled meetings streamlining the entire event lifecycle without manual involvement.
4. **Testing & Reliability Improvements**:
   - Designed and executed test suites to validate scheduling logic, platform integrations, and workflow consistency, ensuring high system reliability across diverse use cases.

Description: Teamicate is an intelligent meeting coordination platform that automates event scheduling based on user availability. With advanced scheduling algorithms, deep calendar integrations, and seamless video conferencing connectivity, Teamicate eliminates back-and-forth communication and enables effortless collaboration across teams and time zones.

Project: Tellen

Technologies: Python, Pandas, NumPy, HTML, JavaScript, Tailwind CSS, PostgreSQL, OpenAI API

Responsibilities:

1. **Financial Data Extraction & Processing**:
   - Engineered advanced PDF data extraction pipelines using *pdfplumber* to parse complex footnote tables from regulatory filings (10-K, annual reports) and convert them into standardized markdown formats for downstream analysis.
2. **Data Analysis & Metric Comparison**:
   - Developed end-to-end preprocessing and analytical workflows using **Pandas** and **NumPy** to compute year-over-year changes in financial disclosures, enabling precise comparison of key metrics and narrative shifts.
3. **LLM-Based Semantic Matching**:
   - Integrated and fine-tuned **OpenAI LLM** workflows to detect nuanced semantic changes in financial footnotes.
   - Designed domain-specific optimization strategies to enhance accuracy, precision, and recall in content-matching tasks.
4. **Structured Output & Reporting**:
   - Implemented structured **JSON-based model outputs** for clean downstream integration.
   - Built visual comparison layers with color-coded annotations to highlight changes, discrepancies, and inconsistencies for auditors and analysts.
5. **Cross-Functional Collaboration & Deployment**:
   - Worked closely with data analysts, auditors, and product teams to operationalize LLM-driven intelligence within financial audit workflows, ensuring seamless adoption and high reliability in production.

Description: Tellen is a financial document intelligence platform built to extract, analyze, and compare tabular and narrative disclosures from regulatory filings. By combining PDF parsing, advanced data transformation, and LLM-powered semantic matching, the system detects year-over-year changes in financial footnotes and presents them through intuitive, color-coded visual comparisons. Tellen streamlines audit processes, improves accuracy, and enhances transparency in financial compliance workflows.